

Israeli Media Framing Tracker

An end-to-end NLP analytics pipeline that quantifies how five Israeli news outlets frame the same political entities and topics. The system pairs Hebrew and English sentiment models with LLM-based emotion and entity enrichment, then surfaces the results through three custom analytical metrics and a five-tab Streamlit dashboard. Portfolio analytics project. Live daily collection via Windows Task Scheduler. The current snapshot covers 750 articles over 16 days; findings illustrate the methodology rather than make definitive claims about Israeli media.

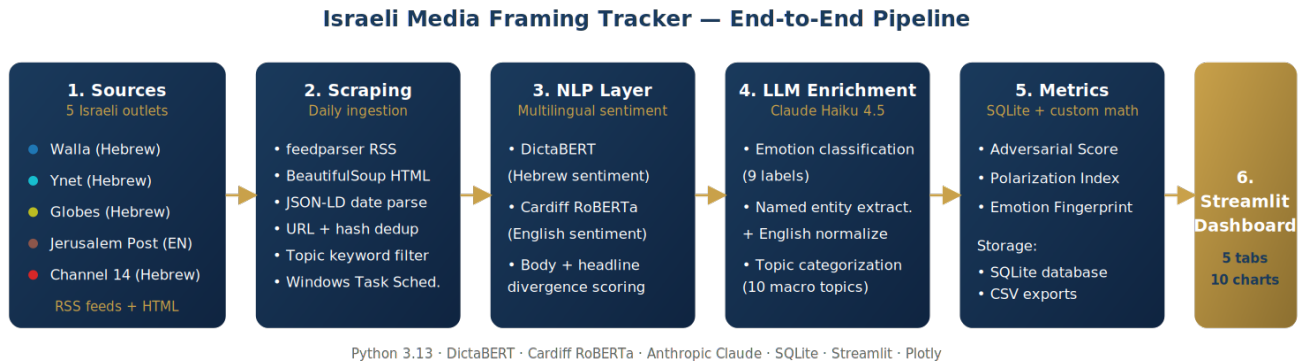


Figure 1. End-to-end pipeline from RSS and HTML ingestion through multilingual NLP, Claude API enrichment, and the final Streamlit dashboard.

The Business Question

Israeli media is polarized, but the diagnosis is usually anecdotal. This project replaces anecdote with measurement and answers one operational question: **When five Israeli outlets cover the same entity or topic, how far apart are their frames, and which camp drives the gap?**

The question splits into four sub-questions an analyst at a PR firm, a corporate communications team, or a political consultancy would need to answer: **(1)** Where is the line? **(2)** Who is the most polarizing public figure? **(3)** Does each outlet carry a stable emotional fingerprint across topics? **(4)** Does political lean explain the variance, or is a second axis at work?

Key Findings

- **Cross-outlet framing gaps are large and structured.** Mean Polarization Index across 13 tracked entities is 0.41 (max 0.75 for Yair Lapid, min 0.05 for Lebanon). The gap correlates with source identity in stable ways, not random noise.
- **Yair Lapid is the most polarizing entity, not Netanyahu.** Walla covers Lapid at -0.667 (hostile), Jerusalem Post at +0.085 (sympathetic). Netanyahu ranks only eighth at 0.27, because every outlet covers him adversarially. The disagreement is over which figures deserve the criticism, not whether to be critical.
- **Each outlet has a stable emotional fingerprint that cuts across topics.** Ynet leads on anger (42%); Walla on anger and fear (36% / 24%); Channel 14 on a distinctive Hopeful register (14% pride, 9% joy); Globes on anticipation and fear (24% / 22%); Jerusalem Post on anger and anticipation (32% / 24%). The signature persists when topic mix is controlled for.
- **Political lean only partly explains framing.** The left-right axis predicts framing on Judicial & Legal (polarization 0.38) and Coalition & Government (0.40), but not on Security Operations or International Diplomacy. A second axis is at work: domestic versus international focus.

Bottom Line for a Stakeholder

For PR and corporate communications. Globes is structurally different from the other four outlets on both emotional register and topic mix, making it the right entry point for any financial or commercial pitch. Ynet's adversarial baseline requires a different playbook.

For political consulting. The Polarization Index ranks public figures by how divided their coverage is, a leading indicator of reputational exposure. A figure with a high score is reading two different narratives about themselves depending on the outlet. The Entity Tracker tab quantifies that gap and shows which outlets drive each side.

Methodology

Adversarial Coverage Score. For each article, `sentiment_intensity` equals `pos_score` minus `neg_score`, then averaged over a group (usually all coverage by one source of one entity). Range -1 (fully adversarial) to +1 (fully sympathetic). Empirically left-skewed in Israeli political coverage; the metric name reflects that observed distribution.

Polarization Index. For an entity or topic, compute the mean Adversarial Score per source (or per political lean). The Polarization Index is the gap between the highest and lowest mean, with a standard-deviation diagnostic reported alongside. Sources with fewer than two articles on the entity are excluded to avoid inflating the index on thin coverage.

Emotion Fingerprint. Each article carries one of nine Claude-assigned emotion labels. Labels collapse into four registers for the radar charts: Hostile (anger, disgust, disappointment), Anxious (fear, sadness, anxiety, tension, concern), Hopeful (anticipation, joy, pride, relief), Neutral (neutral, surprise).

Skills Demonstrated

- **Multilingual NLP pipeline.** DictaBERT for Hebrew (648 articles), Cardiff RoBERTa for English (102 articles), with separate body and headline passes to capture clickbait amplification.
- **LLM enrichment.** Anthropic Claude (Haiku 4.5) for nine-label emotion classification, named entity recognition with Hebrew-to-English normalization (1,002 unique entities), and 12-category topic labeling.
- **Three custom metrics designed from first principles.** Adversarial Coverage Score, Polarization Index, Emotion Fingerprint. Defined, documented, and validated against thin-data edge cases.
- **Five-tab Streamlit dashboard** with ten interactive Plotly visualizations: KPI cards, emotion heatmaps, topic specialization heatmaps, adversarial bar charts, four-axis radars, polarization rankings, and most-adversarial-headlines tables.
- **Production engineering.** JSON-LD date extraction fixed a 35% missing-timestamp bug on Walla and Channel 14; a backfill script then recovered 100% of those dates. Daily ingestion via Windows Task Scheduler, SQLite deduplication, structured logging.
- **Analyst judgment.** Honest scoping, explicit limitations, in-app methodology tab. Findings framed as illustrations of the methodology rather than overclaimed conclusions on a thin window.

Tech Stack

Python 3.13. Hugging Face transformers with DictaBERT and Cardiff RoBERTa. Anthropic Claude API. feedparser, requests, BeautifulSoup, lxml, and JSON-LD parsing for ingestion. SQLite via SQLAlchemy, with CSV exports for portability. Streamlit and Plotly for the interactive dashboard. pandas and numpy for the analytics layer. Windows Task Scheduler for daily runs.